

## Pigeonhole principle:

If  $(k+1)$  or more objects are placed into  $k$  boxes, then there is at least one box containing two or more of the objects.

Ex: if there are 11 players in a soccer team that wins 12-0, there must be one player in the team who scored at least twice.

Generalisation

→ if  $N$  objects are placed in  $k$  boxes, then there is at least one box containing at least  $\lceil \frac{N}{k} \rceil$  of the objects.

Ex:- Let  $S$  contain any 6 positive integers. Then there is a pair of numbers in  $S$  whose difference is divisible by 5.

Sol: Let  $S = \{a_1, a_2, a_3, a_4, a_5, a_6\}$ . Each of these have remainders when divided by 5.

Possible remainders: 0, 1, 2, 3, or 4

6 Numbers      5 remainders

↓ some pair has the same remainder (PHP)

$$a_i = 5m + r \quad a_j = 5n + r$$

$$\begin{aligned} a_i - a_j &= (5m + r) - (5n + r) \\ &= 5(m - n) \end{aligned}$$

which is divisible by 5



# Proving things with the pigeonhole principle -

Step 1:- Decide what the pigeons are?

Step 2:- Set up the pigeonholes.

[ you want to do this so that when you get pigeons in the same pigeonhole they have the property you want ]

Guidelines:

Set things up so that there are fewer pigeonholes than pigeons.

Step 3: Give a rule for assigning the pigeons to the pigeonholes.

Step 4: Apply the PHP to your setup and get the desired conclusion.



# Pigeonhole principle.

→ fundamental but powerful tool in combinatorics.

\* PHP  
When  $n$  pigeons are put into  $k$  pigeonholes. There exist at least 1 pigeonhole containing not less than

$\lceil \frac{n}{k} \rceil$  pigeons and at least one pigeonhole containing not more than

$\lfloor \frac{n}{k} \rfloor$  pigeons

$\lceil x \rceil$ : smallest integer not less than  $x$ .  
 $\lfloor x \rfloor$ : greatest integer not greater than  $x$ .

Proof:-

$$\lceil \pi \rceil = 4, \lfloor \pi \rfloor = 3$$

Assume all pigeonholes contain not more than  $\lceil \frac{n}{k} \rceil - 1$  pigeons.

$$\therefore \lceil x \rceil - 1 < x$$

$$\Rightarrow k \times (\lceil \frac{n}{k} \rceil - 1) < k \times \frac{n}{k} = n \text{ pigeons}$$

contradiction

Assume all pigeonholes contain not more than  $\lfloor \frac{n}{k} \rfloor + 1$  pigeons

$$\therefore \lfloor x \rfloor + 1 > x \Rightarrow k \times (\lfloor \frac{n}{k} \rfloor + 1) > k \times \frac{n}{k} = n \text{ pigeons}$$

contradiction

→ Note: pigeons and pigeonholes are not readily available in usual mathematical problems. Therefore we have to create them suitably in order to adopt the theorem.

Ex 1. 51 numbers are chosen from the integers 1 and 100 inclusively. prove that 2 of the chosen integers are consecutive.

Soln

existence of consecutive. } Pigeonhole.  
main theme

$\{1, 2\} \{3, 4\} \dots \{99, 100\}$

↳ 50 pigeonhole. 51 pigeons.

at least 1 pigeonhole containing

$$\lceil \frac{51}{50} \rceil = 2 \text{ pigeons.}$$



Ex. 2 :-

10 integers are chosen from 1 to 100 inclusively. prove that we can find 2 disjoint non-empty subsets of the chosen integers such that the 2 subsets give the same sum of elements.

Soln:- 2 subsets having same  
sum of elements

↳ Pigeonhole

sum varies from

$$1 \text{ to } 955 \quad \left( \begin{array}{l} 91+92+\dots \\ +100=955 \end{array} \right)$$

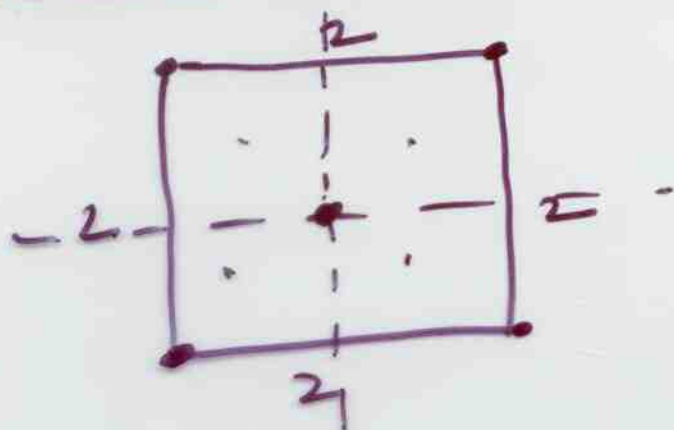
No. of Pigeons =  $2^{10} - 1 = 1023$  pigeons  
(Non-empty subsets)

$$\left\lceil \frac{1023}{955} \right\rceil = 2$$

$\Rightarrow$  2 subsets giving  
the same sum  
of elements



Ex: There are 5 points in a square of side length 2. prove that there exist 2 of them having a distance not more than  $\sqrt{2}$



maximize the  
minimal  
distance.

4 pigeonholes.

5 pigeons.

2 points must be  
in the same  
unit square.

Ex:-

Ex: Show that if any five Numbers from 1 to 8 are chosen. then two of them will add to 9

Soln: Construct four different sets, each containing two numbers that add up to 9 as.

$$A_1 = \{1, 8\}, A_2 = \{2, 7\}, A_3 = \{3, 6\}, A_4 = \{4, 5\}$$

Each of the five numbers chosen must belong to one of these sets. Since there are only four sets, the PMP tells us that two of the chosen numbers belong to the same set.

Extended Pigeonhole principle.

[If No. of pigeons is much larger than the No. of pigeonholes]

If  $n$  pigeons are assigned to  $m$  pigeonholes, then one of the pigeonholes must contain at least  $\lfloor (n-1)/m \rfloor + 1$  pigeons.

Proof: if each pigeonhole contains ~~no~~ no more than  $\lfloor (n-1)/m \rfloor$  pigeons. Then there are at most  $m \cdot \lfloor (n-1)/m \rfloor \leq m \cdot (n-1)/m = n-1$  pigeons.

in all, contradiction so one of the pigeonholes must contain at least  $\lfloor (n-1)/m \rfloor + 1$



Ex: Show that if any 30 people are selected, then one may choose a subset of five so that all five were born on the same day of week.

Soln: 30 pigeons.  
7 pigeonholes.

$\lfloor (30-1)/7 \rfloor + 1$  or 5 of the pigeons must have been born on the same day of week.

Ex: Show that if 30 dictionaries in a library contain a total of 61,327 pages then one of the dictionaries must have at least 2045 pages.

Soln: 61,327 pigeons.  
30 pigeonholes.

$\lfloor 61,326/30 \rfloor + 1$  or 2045 pages





**1. For every 27 word sequence in the US constitution, at least two words will start with the same letter.**

There are 27 words or "pigeons" that can start with one of the 26 different English letters or "pigeonholes." By the pigeonhole principle, two of the words must start with the same letter.

**2. In New York City, there are two non-bald people who have the same number of hairs on their head.**

The human head can contain up to several hundred thousand hairs, with a maximum of about 500,000. In comparison there are millions of people in New York City. Consequently, at least two of them must share the same number of hairs.

**3. Two or more people reading this blog will have the same birthday.**

There are 366 possible birthdays (including February 29 in a leap year) and this blog has many more than 367 readers. Therefore two of you must share the same birthday.

**4. On Thanksgiving, two or more of the consumed turkeys will have the same weight when rounded to the nearest millionth of a pound.**

Turkeys weigh roughly 15 pounds, with the largest recorded at 37 pounds. If we could weigh all turkeys to the millionth of a pound, then we have 37 million possible values for weight. In comparison, there are an estimated 46 million turkeys consumed on Thanksgiving. By the pigeonhole principle, two of those turkeys must have the same weight to the nearest millionth.

**5. At least two WordPress.com blogs have the same number of yearly comments.**

At the time of this writing, there are almost 4.75 million wordpress.com blogs. I would estimate one million as a safe minimum for yearly comments. Thus, the pigeonhole principle implies at least two wordpress.com blogs must have the same number of yearly comments.

**6. In a packed Carnegie Hall performance, there will be two people who have the same first and last initials.**

Each initial can take on one of the 26 English letters, meaning there are  $26 \times 26 = 676$  possible values for the first and last initials. The main hall at Carnegie seats roughly 2,800. Consequently, two of the patrons on a sold out show must share the same first and last initials.

**7. If you pick five cards from a standard deck of 52 cards, then at least two will be of the same suit.**

Each of the five cards can belong to one of four suits. By the pigeonhole principle, two or more must belong to the same suit.

**8. If you have 10 black socks and 10 white socks, and you are picking socks randomly, you will only need to pick 11 socks to find a matching pair.**

The three socks can be one of two colors. By the pigeonhole principle, at least two must be of the same color.

Another way of seeing this is by thinking sock by sock. If the second sock matches the first, then we are done. Otherwise, pick the third sock. Now the first two socks already cover both color cases. The third sock must be one of those and form a matching pair.

